Time Series Analysis & Classification

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Chapter XI: Multivariate Time Series

1. Lagged regression models
2. Multivariate Time Series Regression
3. Vector Autoregressive (VAR) models
4. Graphical Models for Time Series

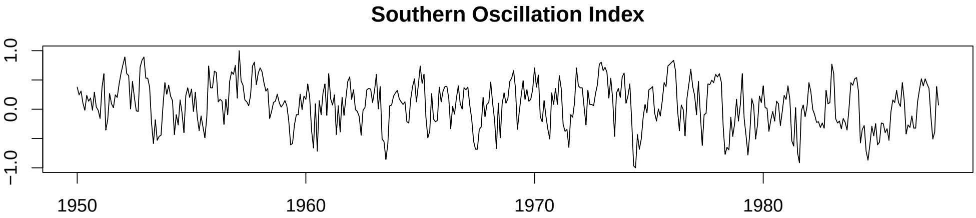
Lagged regression models

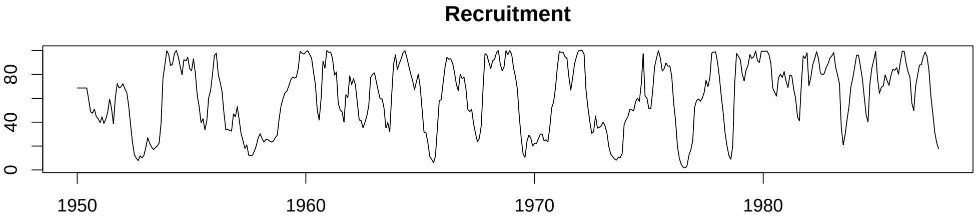
Southern Oscillation Index (SOI) and Recruitment

- Monthly data for 1950-1987 (453 months)
- Southern Oscillation Index (SOI): Measures changes in air pressure, linked to sea surface temperatures in the central Pacific Ocean.
- Recruitment: Number of new fish.
- El Niño effect: Central Pacific warming every 3-7 years, influencing both SOI and Recruitment.

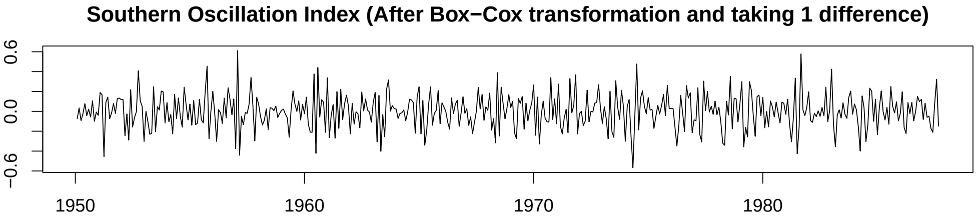
Key takeaway: The SOI and Recruitment series potentially exhibit relationships influenced by the El Niño effect, with implications for environmental and fisheries management.

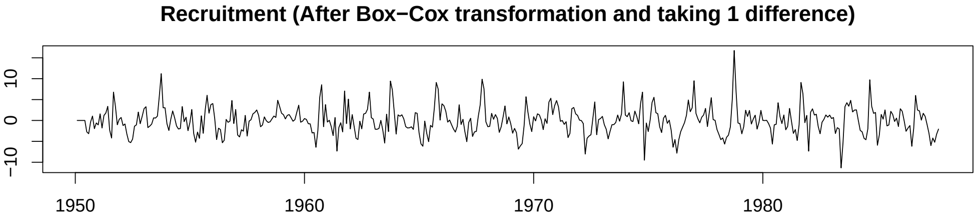
SOI and Recruitment





SOI and Recruitment





Lagged Regression Models

- Consider a lagged regression model of the form:

$$Y_t = \sum_{h=-\infty}^{\infty} \beta_h X_{t-h} + V_t$$

  where:
  - X_t is observed input time series
  - Y_t is observed output time series
  - V_t is stationary noise process

- Useful for:
  - Identifying best linear relationship between time series
  - Forecasting one time series from another

Cross-Covariance, Cross-Spectrum, and Squared Coherence

Cross-Covariance Function

- Recall: The autocovariance function of a stationary process $\{X_t\}$ is defined as:
$$\gamma_X(h) = \mathbb{E}\left[(X_{t+h} - \mu_x)(X_t - \mu_x)\right]$$

- Cross-Covariance: For two jointly stationary processes $\{X_t\}$ and $\{Y_t\}$, the cross-covariance function is:

$$\gamma_{XY}(h) = \mathbb{E}\left[(X_{t+h} - \mu_x)(Y_t - \mu_y)\right]$$

Jointly Stationary Processes

Two time series, say, X_t and Y_t, are said to be jointly stationary if:

- Each of them is stationary individually:
  - Constant means: μ_x, μ_y
  - Auto-covariances depend only on lag h
- Cross-covariance function also depends only on lag h

Cross-Correlation Function (CCF)

- The cross-correlation function of jointly stationary processes $\{X_t\}$ and $\{Y_t\}$ is defined as:

$$\rho_{XY}(h) = \frac{\gamma_{XY}(h)}{\sqrt{\gamma_X(0)\,\gamma_Y(0)}}$$

- Properties:
  - $-1 \leq \rho_{XY}(h) \leq 1$
  - $\rho_{XY}(h) = \rho_{YX}(-h)$
  - In general, $\rho_{XY}(h) \not\equiv \rho_{XY}(-h)$

- Insight: Autocorrelation and cross-correlation functions (ACF and CCF) are useful tools for analyzing the joint behavior of two stationary time series that may be related in an unspecified way.

Example 1: Joint Stationarity

- Consider two time series X_t and Y_t formed from the sum and difference of two successive values of a white noise process:

$$X_t = W_t + W_{t-1}, \quad Y_t = W_t - W_{t-1}$$

where $\{W_t\}$ are independent random variables with mean zero and variance σ_w^2.

- It is easy to show that:

$$\gamma_X(0) = \gamma_Y(0) = 2\sigma_W^2,$$

$$\gamma_X(1) = \gamma_X(-1) = \sigma_W^2, \quad \gamma_Y(1) = \gamma_Y(-1) = -\sigma_W^2$$

Example 1: Joint Stationarity

- Because only one term contributes, the cross-covariance function:

$$\gamma_{XY}(1) = \mathrm{Cov}(X_{t+1}, Y_t) = \mathrm{Cov}(W_{t+1} + W_t, W_t - W_{t-1}) = \sigma_W^2$$

$$\gamma_{XY}(0) = 0, \quad \gamma_{XY}(-1) = -\sigma_W^2$$

- From this, we obtain the cross-correlation function:

$$\rho_{XY}(h) = \begin{cases} 0 & \text{if } h = 0 \text{ or } |h| \geq 2 \\ 1/2 & \text{if } h = 1 \\ -1/2 & \text{if } h = -1 \\ 0 & \text{if } |h| \geq 2 \end{cases}$$

- Conclusion: The autocovariance and cross-covariance functions depend only on the lag h, so the series $\{X_t\}$ and $\{Y_t\}$ are jointly stationary.

Example 2: Joint Stationarity

- Suppose $Y_t = \beta X_{t-\ell} + W_t$, where:
  - $\{X_t\}$ is stationary
  - $\{W_t\}$ is white noise, zero-mean, and uncorrelated with $\{X_t\}$
- Then:
  - $\{X_t\}$ and $\{Y_t\}$ are jointly stationary
  - $\mu_Y = \beta \mu_X$
  - $\gamma_{XY}(h) = \beta \gamma_X(h + \ell)$
- Interpretation:
  - If $\ell > 0$, we say X_t leads Y_t
  - If $\ell < 0$, we say X_t lags Y_t

Sample Cross-Covariance and Cross-Correlation Functions

- Sample cross-covariance:

$$\hat{\gamma}_{XY}(h) = \frac{1}{n}\sum_{i=1}^{n-h}(X_{t+h} - \bar{X})(Y_t - \bar{Y})$$

for $h \geq 0$ and $\hat{\gamma}_{XY}(h) = \hat{\gamma}_{YX}(-h)$ for $h < 0$

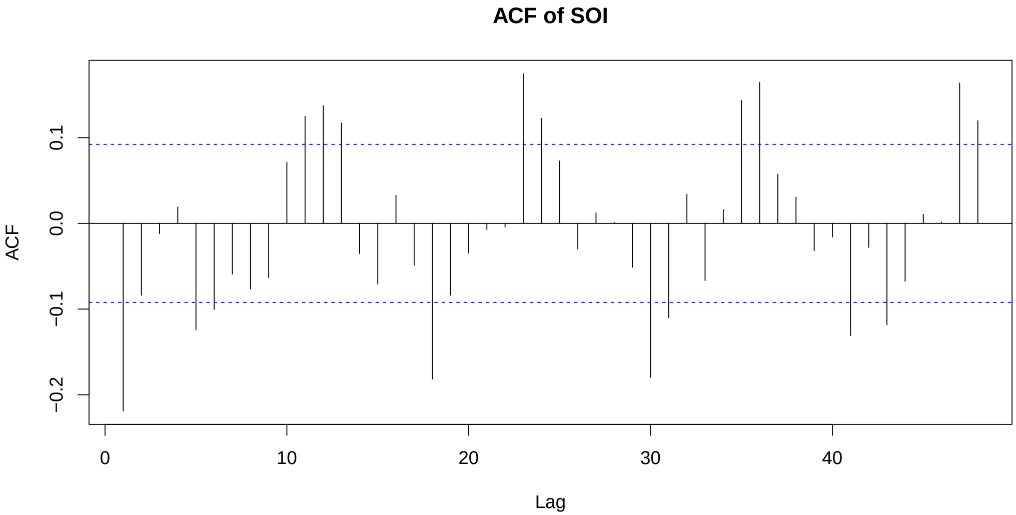
- Sample cross-correlation functions (CCF):

$$\hat{\rho}_{XY}(h) = \frac{\hat{\gamma}_{XY}(h)}{\sqrt{\hat{\gamma}_X(0)\hat{\gamma}_Y(0)}}$$

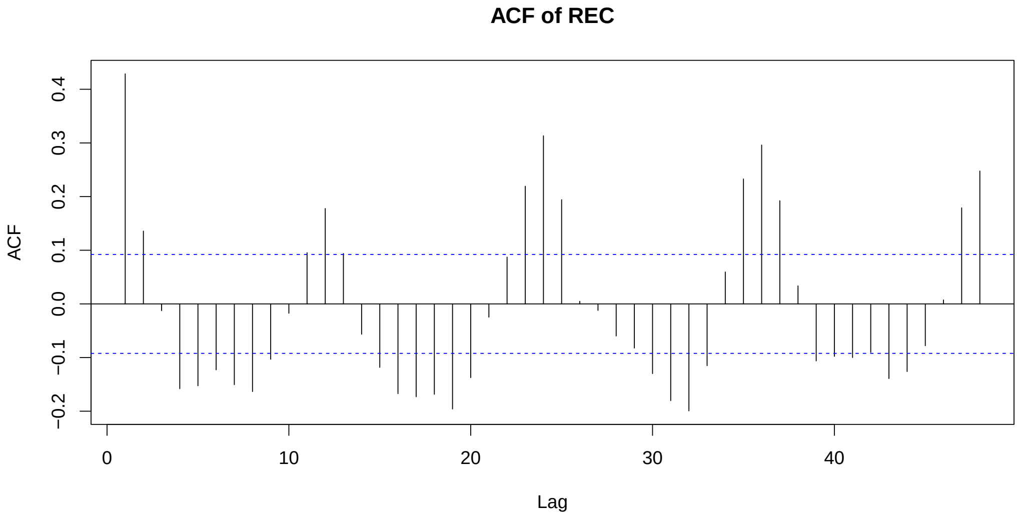
Sample Cross-Covariance and Cross-Correlation Functions

- If either series is white: $\hat{\rho}_{XY}(h) \sim \mathcal{AN}(0, 1/\sqrt{n})$
- We can look for peaks in the sample CCF to identify a leading or lagging relation. (Recall that the ACF of the input series peaks at $h = 0$.)

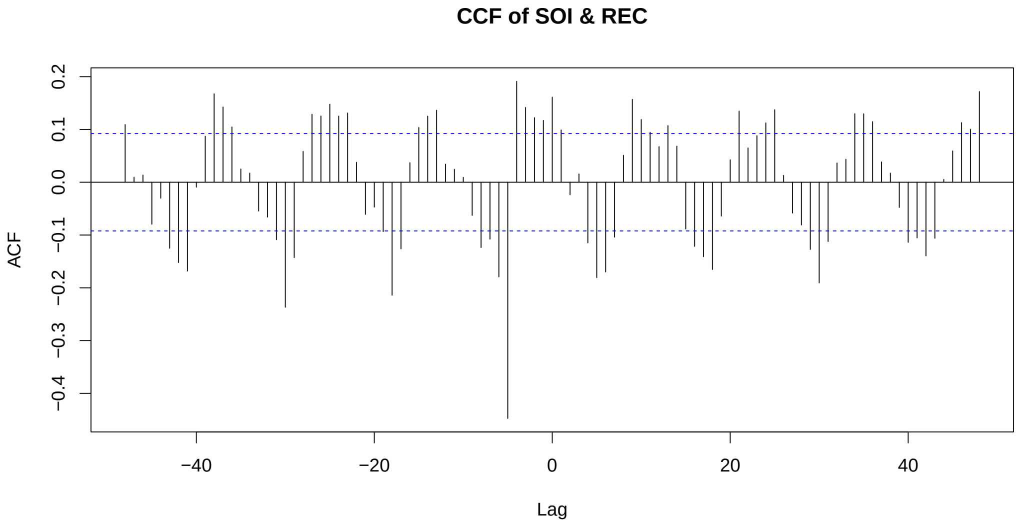
SOI and Recruitment



SOI and Recruitment



SOI and Recruitment



SOI and Recruitment

ACF Insight:

- ACFs of both series show periodicity with strong positive correlation at lags of 12, 24, 36, ... months (annual cycle).
- Negative correlation appears at lag 6, suggesting inverse behavior six months apart.
- This pattern resembles a sinusoidal signal with a 12-month period.

SOI and Recruitment

CCF Insight:

- The cross-correlation function peaks at lag $h = -6$, indicating that SOI leads Recruitment by six months.
- The negative sign of the correlation implies an inverse relationship: increases in SOI are associated with decreases in Recruitment.

Conclusion: SOI may help predict Recruitment with a lead time of six months, though the relationship is inverse.

Cross-Spectrum

To analyze lagged regression in the frequency domain, we'll need the notion of coherence, the analog of cross-correlation in the frequency domain.

- Define the cross-spectrum as the Fourier transform of the cross-covariance:

$$S_{XY}(\nu) = \sum_{h=-\infty}^{\infty} \gamma_{XY}(h) e^{-2\pi i \nu h}$$

$$\gamma_{XY}(h) = \int_{-1/2}^{1/2} S_{XY}(\nu) e^{2\pi i \nu h} d\nu$$

- Properties:
    - $S_{XY}(\nu)$ can be complex-valued
    - $\gamma_{YX}(h) = \gamma_{XY}(-h)$ implies $S_{YX}(\nu) = S_{XY}(\nu)^*$

Squared Coherence Function

- The squared coherence function (sometimes called coherency) is defined as:
$$\rho^2_{XY}(\nu) = \frac{|S_{XY}(\nu)|^2}{S_X(\nu)S_Y(\nu)},$$
where
$$0 \leq \rho^2_{XY}(\nu) \leq 1$$
- Interpretation: Squared coherence at frequency ν represents contribution to squared correlation at that frequency. In other words, it gives a measure of the association or correlation between the input and output series as a function of frequency.
- Recall: The interpretation of spectral density at a frequency ν as the contribution to variance at that frequency.

Example

- We want compute the cross-spectrum between $\{X_t\}$ and the three-point moving average (we assume $\{X_t\}$ is a stationary process with spectral density $S_X(\nu)$).

$$Y_t = \frac{X_{t-1} + X_t + X_{t+1}}{3}$$

- First, compute the cross-covariance:

$$\begin{aligned}
\gamma_{XY}(h) = \mathrm{Cov}(X_{t+h}, Y_t) &= \frac{1}{3}\,\mathrm{Cov}(X_{t+h}, X_{t-1} + X_t + X_{t+1}) \\
&= \frac{1}{3}[\gamma_X(h+1) + \gamma_X(h) + \gamma_X(h-1)]
\end{aligned}$$

- By the spectral representation:

$$\gamma_{XY}(h) = \int_{-1/2}^{1/2} \frac{1 + 2\cos(2\pi\nu)}{3} S_X(\nu) e^{2\pi i\nu h}\, d\nu$$

Example

- Using the uniqueness of the Fourier transform, we obtain the cross-spectrum as:

$$S_{XY}(\nu) = \frac{1 + 2\cos(2\pi\nu)}{3} S_X(\nu), \qquad \text{which is real-valued.}$$

- We have also, the spectral density of Y_t:

$$\begin{aligned} S_Y(\nu) &= \frac{1}{9}\left[1 + 4\cos(2\pi\nu) + 2\cos(4\pi\nu)\right] S_X(\nu) \\ &= \frac{1}{9}\left[1 + 2\cos(2\pi\nu)\right]^2 S_X(\nu) \end{aligned}$$

- Hence, the squared coherence between X_t and Y_t is:

$$\rho^2_{XY}(\nu) = 1 \quad \text{for all } \nu$$

Perfect Coherence

- Perfect coherence occurs when:

$$\rho^2_{XY}(\nu) = 1 \quad \text{for all } \nu$$

  Perfect coherence implies that Y_t is a perfectly filtered version of X_t at all frequencies.

- Implications:
  - The cross-spectrum $S_{XY}(\nu)$ fully explains the relationship between X_t and Y_t at all frequencies.
  - There is no frequency at which Y_t contains any component not linearly related to X_t
  - Y_t is a deterministic, linear transformation (filter) of X_t.

- Conclusion: When coherence is 1 at all frequencies, Y_t contains no independent noise beyond what is present in X_t. Hence,

$$Y_t = (\text{linear filter of})\ X_t$$

Estimating Squared Coherence

- Recall that we estimated the spectral density using the squared modulus of the DFT of the series,:

$$\hat{S}_X(\nu_k) = |X(\nu_k)|^2$$

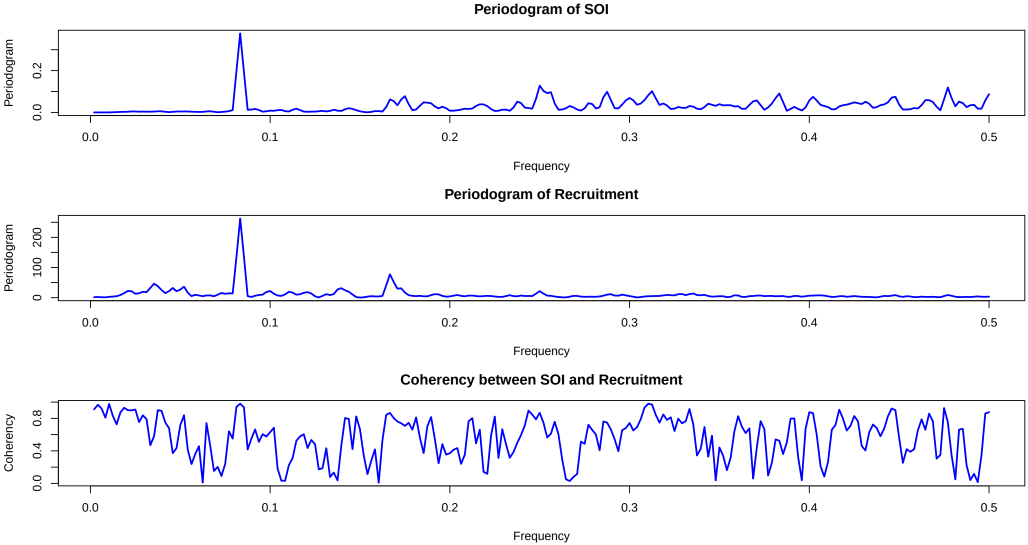
- We can estimate the cross spectral density using the same sample estimate:

$$\hat{S}_{XY}(\nu_k) = X(\nu_k)\,Y(\nu_k)^*$$

  where $Y(\cdot)^*$ is the complex conjugate of the DFT of Y_t
- We can, thus, estimate the squared coherence using these estimates,:

$$\hat{\rho}^2_{XY}(\nu) = \frac{|\hat{S}_{XY}(\nu)|^2}{\hat{S}_X(\nu)\hat{S}_Y(\nu)}$$

SOI and Recruitment Spectra



Decibels in Spectral Density

- Spectral density can vary over several orders of magnitude.
- Using the decibel (dB) scale makes large dynamic ranges more interpretable and easier to visualize.
- The decibel scale is defined as:

$$\text{Power in dB} = 10 \log_{10}(\text{Power})$$

- Log-scaling compresses high values and expands low values - useful when analyzing both strong and weak spectral components on the same plot.
- Decibels are common in engineering and signal processing for comparing signal strengths or power ratios.
- Peaks and differences in spectral density become more visually prominent and interpretable in the dB scale.

SOI and Recruitment Spectra

Lagged Regression Models in the Frequency Domain

Lagged Regression Model in the Frequency Domain

- Recall the lagged regression model of the form:

$$Y_t = \sum_{j=-\infty}^{\infty} \beta_j X_{t-j} + V_t,$$

  where, X_t is the observed input time series, Y_t is the observed output time series, V_t is a stationary noise process.

- Goal: Estimate the coefficients β_j that characterize the linear dependence of Y_t on past, present, and future values of X_t.

- This model is often used to study how the input series influences the output over time, allowing for delayed or anticipatory effects through the lag structure in β_j.

Projection Theorem and Optimal Coefficients

- The projection theorem tells us that the coefficients $\{\beta_j\}$ that minimize the mean squared error:

$$\mathbb{E}\left[\left(Y_t - \sum_{j=-\infty}^{\infty} \beta_j X_{t-j}\right)^2\right]$$

must satisfy the following orthogonality conditions:

$$\mathbb{E}\left[\left(Y_t - \sum_{j=-\infty}^{\infty} \beta_j X_{t-j}\right) X_{t-k}\right] = 0, \quad \text{for all } k \in \mathbb{Z}$$

- These conditions imply the convolution equations:

$$\sum_{j=-\infty}^{\infty} \beta_j \gamma_X(k-j) = \gamma_{XY}(k), \quad \text{for all } k \in \mathbb{Z}$$

Solving for β_j Using Spectral Methods

- We could solve the orthogonality equations for the coefficients β_j using: the sample autocovariance function $\hat{\gamma}_X(h)$, and the sample cross-covariance function $\hat{\gamma}_{XY}(h)$.

- However, it is often more convenient to use the estimates of the spectral density $S_X(\nu)$, and the cross-spectral $S_{XY}(\nu)$ density functions, since the convolution with $\{\beta_j\}$ in the time domain is equivalent to multiplication by the Fourier transform of $\{\beta_j\}$ in the frequency domain.

Solving for β_j Using Spectral Methods

- From the orthogonality conditions, we obtain for $k = 0, \pm 1, \pm 2, \ldots$:

$$\int_{-1/2}^{1/2} \left(\sum_{j=-\infty}^{\infty} \beta_j e^{2\pi i \nu (k-j)} \right) S_X(\nu)\, d\nu = \int_{-1/2}^{1/2} e^{2\pi i \nu k} S_{YX}(\nu)\, d\nu$$

- Define the Fourier transform of the coefficient sequence $\{\beta_j\}$ as:

$$B(\nu) = \sum_{j=-\infty}^{\infty} \beta_j e^{-2\pi i \nu j}$$

- Then the orthogonality condition becomes:

$$\int_{-1/2}^{1/2} e^{2\pi i \nu k} B(\nu) S_X(\nu)\, d\nu = \int_{-1/2}^{1/2} e^{2\pi i \nu k} S_{XY}(\nu)\, d\nu$$

- By uniqueness of the Fourier transform:

$$B(\nu) S_X(\nu) = S_{XY}(\nu)$$

Estimating β_j in the Frequency Domain

- We can estimate the transfer function $B(\nu)$ as:

$$\hat{B}(\nu_k) = \frac{\hat{S}_{XY}(\nu_k)}{\hat{S}_X(\nu_k)}$$

- To recover the filter coefficients $\hat{\beta}_j$, approximate the inverse Fourier transform:

$$\beta_j = \int_{-1/2}^{1/2} e^{2\pi i \nu j} B(\nu)\, d\nu,$$

  using the discrete Fourier approximation:

$$\hat{\beta}_j = \frac{1}{m} \sum_{k=0}^{m-1} \hat{B}(\nu_k) e^{2\pi i \nu_k j}$$

Summary

Here is the approach:

1. Estimate the spectral density and cross-spectral density.

2. Compute the transfer function $\hat{B}(\nu)$.

3. Take the inverse Fourier transform to obtain the impulse response function β_j.

Considerations on Representations

It is often useful to consider both representations:

$$Y_t = \sum_{j=-\infty}^{\infty} \alpha_j X_{t-j},$$

$$X_t = \sum_{j=-\infty}^{\infty} \beta_j Y_{t-j},$$

since there might be a more parsimonious representation in terms of one than the other. (Just as a small AR model often cannot be well approximated by a small MA model.)

SOI and Recruitment: Input-Output Relationship

- The high coherence between the SOI and Recruitment series noted previously suggests a lagged regression relation between the two series.
- A natural direction in this situation is implied because we feel that the sea surface temperature should be the input and the Recruitment series should be the output.
- Although we think naturally of the SOI as the input and the Recruitment as the output, two input-output configurations are of interest.

SOI and Recruitment: Input-Output Relationship

- With this in mind, let X_t be the SOI series and Y_t the Recruitment series.
- With SOI as the input, the model is (W_t and V_t are white noise processes) :

$$Y_t = \sum_{r=-\infty}^{\infty} a_r X_{t-r} + W_t$$

whereas a model that reverses the two roles would be:

$$X_t = \sum_{r=-\infty}^{\infty} b_r Y_{t-r} + V_t$$

- Even though there is no plausible environmental explanation for the second model (Recruitment as input, SOI as output), considering both possibilities helps in identifying a parsimonious transfer function model.

SOI as the input

- Based on a lagged regression analysis, the estimated impulse response function for SOI (as input) shows notable coefficients at lags 5 through 8:

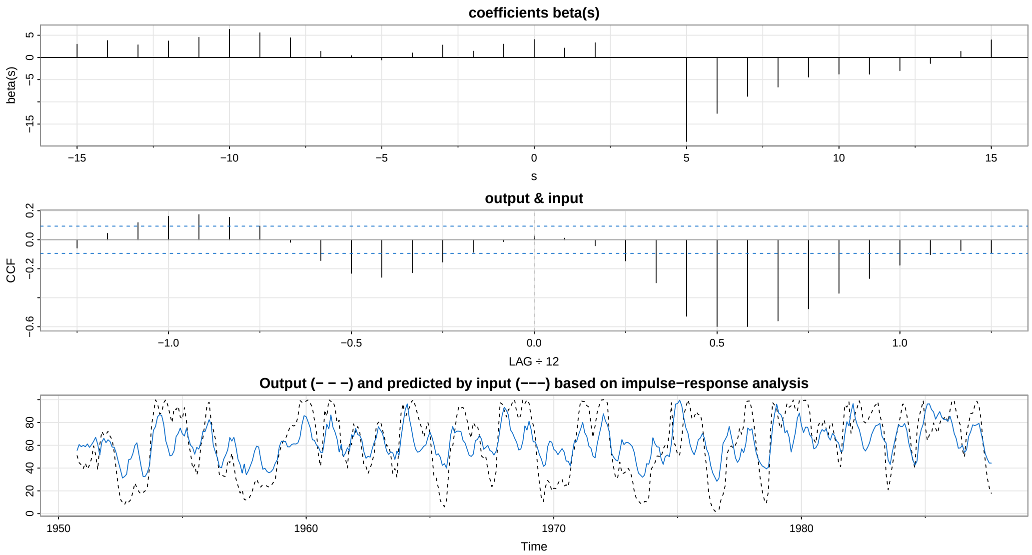
$$\beta_5 = -18.5, \quad \beta_6 = -12.3, \quad \beta_7 = -8.5, \quad \beta_8 = -7.0$$

- The resulting prediction equation is:

$$Y_t = 66 - 18.5X_{t-5} - 12.3X_{t-6} - 8.5X_{t-7} - 7.0X_{t-8} + W_t$$

- The coefficients exhibit an approximately exponential decay after lag 5, suggesting a simple transfer function model structure.

SOI as input


coefficients beta(s)
beta(s)
s
output & input
CCF
LAG ÷ 12
Output (– – –) and predicted by input (–––) based on impulse−response analysis
Time

Inverse Regression: Recruitment as Input

- When reversing the roles and treating Recruitment as the input and SOI as the output, a much simpler regression model results. Only two significant coefficients are identified:

$$\beta_4 = 0.016, \quad \beta_5 = -0.02$$

- The corresponding prediction equation is:

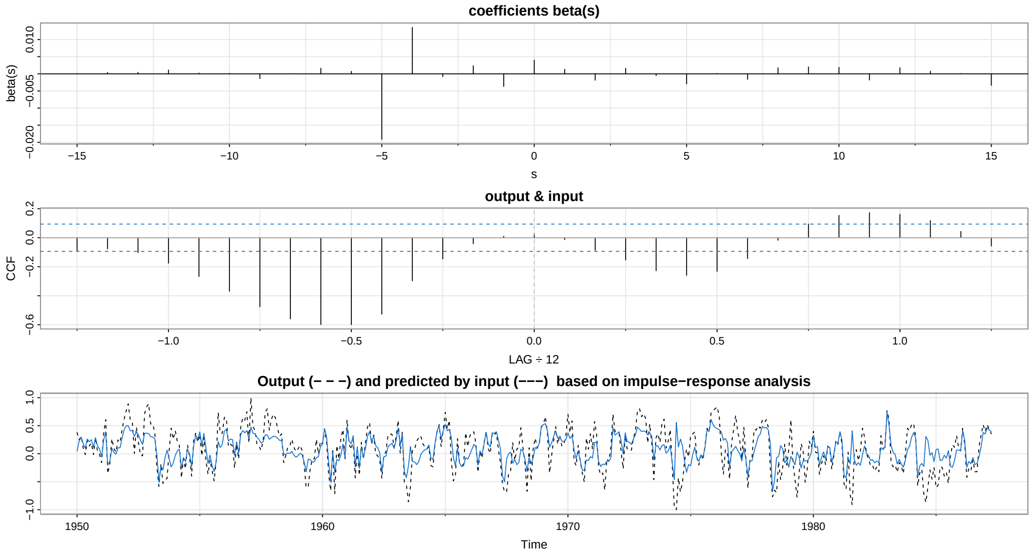
$$X_t = 0.41 + 0.016\, Y_{t+4} - 0.02\, Y_{t+5} + v_t$$

- Multiplying both sides by $50B^5$ and rearranging yields a parsimonious transfer function model:

$$(1 - 0.8B)\, Y_t = 20.5 - 50B^5 X_t + \varepsilon_t$$

where ε_t is a white noise process.

Recruitment as input



Multivariate Time Series Regression

Multivariate Time Series Regression: Overview

- In time series analysis, we often encounter more than one time-dependent response (input) variable.
- Goal: Extend univariate regression models to accommodate multiple dependent variables over time.
- Suppose at each time t we observe k outputs:

$$\mathbf{y}_t = (y_{t1}, y_{t2}, \ldots, y_{tk})^\top$$

and r predictors or input variables:

$$\mathbf{z}_t = (z_{t1}, z_{t2}, \ldots, z_{tr})^\top$$

Multivariate Time Series Regression: Overview

- Each output is modeled as a linear function of the predictors:

$$y_{ti} = \beta_{i1} z_{t1} + \beta_{i2} z_{t2} + \cdots + \beta_{ir} z_{tr} + w_{ti}, \quad i = 1, \ldots, k$$

- The errors w_{ti} satisfy: $\mathrm{Cov}(w_{si}, w_{tj}) = \sigma_{ij}^2$ if $s = t$, and 0 otherwise. That is the errors:
    - are uncorrelated across time,
    - may be correlated across output variables at the same time t.

- This reflects shared random effects or measurement errors across series.

Matrix Form of the Multivariate Regression Model

- We can express the full model compactly in matrix form:

$$\mathbf{y}_t = B\mathbf{z}_t + \mathbf{w}_t$$

  - $\mathbf{y}_t$: $k \times 1$ vector of output variables
  - $\mathbf{z}_t$: $r \times 1$ vector of input variables
  - B: $k \times r$ matrix of regression coefficients (β_{ij})
  - $\mathbf{w}_t$: $k \times 1$ vector of white noise errors

- Assume that the noise vectors $\mathbf{w}_t$:

$$\mathbb{E}[\mathbf{w}_t] = 0, \quad \mathbb{E}[\mathbf{w}_t \mathbf{w}_t^\top] = \Sigma_w$$

  where Σ_w is a $k \times k$ positive definite covariance matrix.

- This allows for correlation between different output variables at each time point (cross-sectional dependence), while maintaining independence over time.

Estimation and Inference

- Let $Y^\top = [\mathbf{y}_1, \ldots, \mathbf{y}_n]$ and $Z^\top = [\mathbf{z}_1, \ldots, \mathbf{z}_n]$. The maximum likelihood estimator, under the assumption of normality, for the regression matrix in this case is
$$\hat{B} = Y^\top Z(Z^\top Z)^{-1}$$
- Estimated error covariance matrix:
$$\hat{\Sigma}_w = \frac{1}{n-r}\sum_{t=1}^{n}(\mathbf{y}_t - \hat{B}\mathbf{z}_t)(\mathbf{y}_t - \hat{B}\mathbf{z}_t)^\top$$
- Standard errors: For each coefficient $\hat{\beta}_{ij}$, $\quad se\,(\hat{\beta}_{ij}) = \sqrt{c_{ii}\,\hat{\sigma}_{jj}}$ where:
  - c_{ii} is the i-th diagonal entry of $(\sum \mathbf{z}_t\mathbf{z}_t^\top)^{-1}$,
  - $\hat{\sigma}_{jj}$ is the j-th diagonal of $\hat{\Sigma}_w$.

  These standard errors help us assess the statistical significance.

Model Selection Criteria

- In multivariate regression, model complexity increases quickly with k (outputs) and r (predictors).

- Akaike Information Criterion (AIC):

$$\mathrm{AIC} = \log|\hat{\Sigma}_w| + \frac{2}{n}(kr + k(k+1))$$

- Bayesian Information Criterion (BIC):

$$\mathrm{BIC} = \log|\hat{\Sigma}_w| + \frac{1}{n}K\log n, \quad K = kr + \frac{k(k+1)}{2}$$

- Interpretation:
  - $\log|\hat{\Sigma}_w|$: fit of the model (log-likelihood term),
  - penalty terms: discourage overfitting by penalizing model size.

  These criteria help balance goodness-of-fit with model complexity.

Vector Autoregressive (VAR) models

Multivariate Time Series: Motivation

- Many applications involve multiple time series recorded simultaneously.
- We denote the observed m-variate time series as:

$$\mathbf{X}_t = (X_{t1}, X_{t2}, \ldots, X_{tm})^T, \quad t \in \mathbb{Z}$$

- Goal: Model dynamic relationships among components of $\mathbf{X}_t$, and forecast future values.
- Extending univariate $ARMA$ models to multivariate setting is not straightforward.
- Vector autoregressive (VAR) models provide a flexible and tractable multivariate extension.

VAR models Applications

- Macroeconomics: Modeling relationships between GDP, inflation, interest rates.
- Finance: Joint modeling of asset returns, volatility spillovers.
- Marketing: Studying the impact of price, advertising, and sales over time.
- Engineering: Multivariate control systems and sensor fusion.

Key Benefit: VAR models do not impose strict a priori causal ordering, data-driven inference of interactions is possible.

Multivariate White Noise

- Definition: A multivariate process $\{\mathbf{w}_t\}$ is said to be white noise with mean zero and covariance matrix $\mathbf{\Sigma}$, denoted $WN(\mathbf{0}, \mathbf{\Sigma})$, if:
  - $\mathbb{E}[\mathbf{w}_t] = \mathbf{0}$
  - $\mathrm{Cov}(\mathbf{w}_{t+h}, \mathbf{w}_t) = \begin{cases} \mathbf{\Sigma}, & h = 0 \\ \mathbf{0}, & h \neq 0 \end{cases}$
- Discussion: Multivariate white noise extends the univariate concept to vector-valued processes. It is the building block for modeling innovation terms in vector autoregressive models (VARs). The lack of autocorrelation (at all non-zero lags) ensures no linear predictability, making $\mathbf{w}_t$ a suitable innovation or error term in time series models.

$VAR(p)$ Model

- A k-dimensional time series $\{\boldsymbol{z}_t\}$ follows a $VAR(p)$ model if:

$$\boldsymbol{z}_t = \boldsymbol{\phi}_0 + \sum_{i=1}^{p} \boldsymbol{\Phi}_i \boldsymbol{z}_{t-i} + \boldsymbol{a}_t,$$

  where:

  - $\boldsymbol{\phi}_0$ is a k-dimensional intercept vector,
  - $\boldsymbol{\Phi}_i$ are $k \times k$ coefficient matrices,
  - $\boldsymbol{a}_t \sim WN(0, \boldsymbol{\Sigma}_a)$.

Remarks:

- VAR models extend univariate AR models to multivariate settings.
- Inference methods from multivariate multiple linear regression apply.
- $VAR(p)$ is a special case of $VARMA(p, q)$ with $q = 0$.

$VAR(1)$ Model

- In this section we will mainly discuss the $VAR(1)$ defined as:

$$\boldsymbol{z}_t = \boldsymbol{\phi}_0 + \boldsymbol{\Phi}_1 \boldsymbol{z}_{t-1} + \boldsymbol{a}_t,$$

- Example: Bivariate $VAR(1)$ model:

$$\text{with } \boldsymbol{z}_t = \begin{bmatrix} z_{1t} \\ z_{2t} \end{bmatrix}, \quad \boldsymbol{\phi}_0 = \begin{bmatrix} \phi_{10} \\ \phi_{20} \end{bmatrix}, \quad \boldsymbol{\Phi}_1 = \begin{bmatrix} \phi_{11} & \phi_{12} \\ \phi_{21} & \phi_{22} \end{bmatrix}.$$

Explicitly:

$$\begin{aligned} z_{1t} &= \phi_{10} + \phi_{11} z_{1,t-1} + \phi_{12} z_{2,t-1} + a_{1t}, \\ z_{2t} &= \phi_{20} + \phi_{21} z_{1,t-1} + \phi_{22} z_{2,t-1} + a_{2t}. \end{aligned}$$

Interpreting Coefficients

- ϕ_{12}: Influence of $z_{2,t-1}$ on z_{1t}.
- ϕ_{21}: Influence of $z_{1,t-1}$ on z_{2t}.
- Special cases:
  - $\phi_{12} = \phi_{21} = 0$: Uncoupled systems.
  - $\phi_{12} = 0$, $\phi_{21} \neq 0$: Unidirectional influence (transfer function structure).
- Application: Control engineering and economics often use such unidirectional dynamics.

Granger Causality

- Intuition
  - A variable z_{1t} Granger-causes z_{2t} if past values of z_{1t} contain information that helps predict z_{2t} beyond what is already in the past of z_{2t}.
  - It is a predictive concept, not a statement about true causal mechanisms.

- Formal Definition:

$$
\begin{aligned}
z_{1t} &= \phi_{10} + \phi_{1,11} z_{1,t-1} + \phi_{1,12} z_{2,t-1} + a_{1t}, \\
z_{2t} &= \phi_{20} + \phi_{1,21} z_{1,t-1} + \phi_{1,22} z_{2,t-1} + a_{2t}.
\end{aligned}
$$

  - If $\phi_{1,21} \neq 0$, then z_{1t} Granger-causes z_{2t}.
  - If $\phi_{1,12} = 0$, then z_{2t} does not Granger-cause z_{1t}.

Stationarity of VAR(1)

- Assume:

$$\boldsymbol{z}_t = \boldsymbol{\Phi}_1 \boldsymbol{z}_{t-1} + \boldsymbol{a}_t, \quad \boldsymbol{\phi}_0 = 0.$$

- Suppose that the time series started at time $t = v$ with initial value $\mathbf{z}_v$, where v is a fixed time point. Using iterated expansion, we get :

$$\boldsymbol{z}_t = \boldsymbol{\Phi}_1^{t-v} \boldsymbol{z}_v + \sum_{i=0}^{t-1} \boldsymbol{\Phi}_1^i \boldsymbol{a}_{t-i}.$$

- Condition: For stationarity, $\boldsymbol{\Phi}_1^{t-v} \boldsymbol{z}_v \to 0$ as $v \to -\infty$.

- Result: All eigenvalues of $\boldsymbol{\Phi}_1$ must lie within the unit circle.

$$|\lambda_j| < 1, \quad \text{for all } j.$$

Stationarity Example

- Consider:

$$\boldsymbol{z}_t = \begin{bmatrix} 5 \\ 3 \end{bmatrix} + \begin{bmatrix} 0.2 & -0.6 \\ 0.3 & 1.1 \end{bmatrix} \boldsymbol{z}_{t-1} + \boldsymbol{a}_t, \quad \boldsymbol{a}_t \sim WN(0, \boldsymbol{\Sigma}_a),$$

$$\boldsymbol{\Sigma}_a = \begin{bmatrix} 1.0 & 0.8 \\ 0.8 & 2.0 \end{bmatrix}.$$

- Eigenvalues of $\boldsymbol{\Phi}_1$: 0.5 and 0.8 $\Rightarrow$ stationary.
-
- Note: Although $\phi_{22} = 1.1 > 1$, this does not violate stationarity.

Moment Equations in VAR(1)

- Assume $\boldsymbol{z}_t$ is stationary. Then:

$$\boldsymbol{\mu} = \mathbb{E}[\boldsymbol{z}_t] = (\boldsymbol{I}_k - \boldsymbol{\Phi}_1)^{-1}\boldsymbol{\phi}_0.$$

- Let $\tilde{\boldsymbol{z}}_t = \boldsymbol{z}_t - \boldsymbol{\mu}$:

$$\tilde{\boldsymbol{z}}_t = \boldsymbol{\Phi}_1 \tilde{\boldsymbol{z}}_{t-1} + \boldsymbol{a}_t.$$

- Taking expectations of $\tilde{\boldsymbol{z}}_t \tilde{\boldsymbol{z}}_t^\top$, we have the covariance matrix:

$$\boldsymbol{\Gamma}_0 = \mathbb{E}[\tilde{\boldsymbol{z}}_t \tilde{\boldsymbol{z}}_t^\top] = \mathbb{E}[\boldsymbol{\Phi}_1 \tilde{\boldsymbol{z}}_{t-1} \tilde{\boldsymbol{z}}_{t-1}^\top \boldsymbol{\Phi}_1^\top] + \mathbb{E}[a_t a_t^\top].$$

  Since a_t is uncorrelated with $\tilde{\boldsymbol{z}}_{t-1}$:

$$\boldsymbol{\Gamma}_0 = \boldsymbol{\Phi}_1 \boldsymbol{\Gamma}_0 \boldsymbol{\Phi}_1^\top + \Sigma_a.$$

- This equation links the stationary covariance matrix $\boldsymbol{\Gamma}_0$ with model parameters $\boldsymbol{\Phi}_1$ and Σ_a.

Solving for the Covariance Matrix Γ_0

- We vectorise the system to have:

$$\mathrm{vec}(\boldsymbol{\Gamma}_0) = (\boldsymbol{\Phi}_1 \otimes \boldsymbol{\Phi}_1)\mathrm{vec}(\boldsymbol{\Gamma}_0) + \mathrm{vec}(\boldsymbol{\Sigma}_a).$$

$$\Rightarrow \mathrm{vec}(\boldsymbol{\Gamma}_0) = (\boldsymbol{I}_{k^2} - \boldsymbol{\Phi}_1 \otimes \boldsymbol{\Phi}_1)^{-1}\mathrm{vec}(\boldsymbol{\Sigma}_a),$$

  where $\mathrm{vec}(\boldsymbol{A})$ is the $k^2 \times 1$ column vector obtained by stacking the columns of A on top of one another, and $A \otimes B$ is the Kronecker product.

- Interpretation:
  - The above equation can be solved for $\boldsymbol{\Gamma}_0$ using known quantities $\boldsymbol{\Phi}_1$ and Σ_a.
  - This equation is linear in $\mathrm{vec}(\boldsymbol{\Gamma}_0)$ and can be efficiently solved numerically.

Multivariate Yule-Walker Equations

- Next, for any positive integer ℓ, multiplying $\tilde{\mathbf{z}}_{t-\ell}$ to the mean-adjusted process equation and taking expectations, we obtain:

$$\boldsymbol{\Gamma}_\ell = \mathbb{E}[\tilde{\mathbf{z}}_t \tilde{\mathbf{z}}_{t-\ell}^\top] = \mathbb{E}[\boldsymbol{\Phi}_1 \tilde{\mathbf{z}}_{t-1} \tilde{\mathbf{z}}_{t-\ell}^\top + \mathbf{a}_t \tilde{\mathbf{z}}_{t-\ell}^\top].$$

- Since $\mathbf{a}_t$ is white noise and uncorrelated with $\tilde{\mathbf{z}}_{t-\ell}$ for $\ell > 0$, the cross-term vanishes:

$$\boldsymbol{\Gamma}_\ell = \boldsymbol{\Phi}_1 \boldsymbol{\Gamma}_{\ell-1}, \quad \ell > 0.$$

- This recursive relation is known as the multivariate Yule-Walker equation.

Multivariate Yule-Walker Equations

- Recursive use:
  - Compute higher-lag autocovariances $\mathbf{\Gamma}_h$.
  - Estimate $\mathbf{\Phi}_1$ as $\mathbf{\Gamma}_1\mathbf{\Gamma}_0^{-1}$.
- Cross-correlation matrices:

$$\rho_{ij}(h) = \frac{\gamma_{ij}(h)}{\sqrt{\gamma_{ii}(0)\gamma_{jj}(0)}}$$

where $(\mathbf{\Gamma}_h)_{ij} = \gamma_{ij}(h)$

Parameter Estimation

- Define:

$$\mathbf{y} = \mathbf{z}_t, \quad \mathbf{x} = \begin{bmatrix} 1 \\ \mathbf{z}_{t-1} \end{bmatrix}, \quad B = \begin{bmatrix} \boldsymbol{\alpha} & \Phi \end{bmatrix}$$

- Write in the Multivariate regression form:

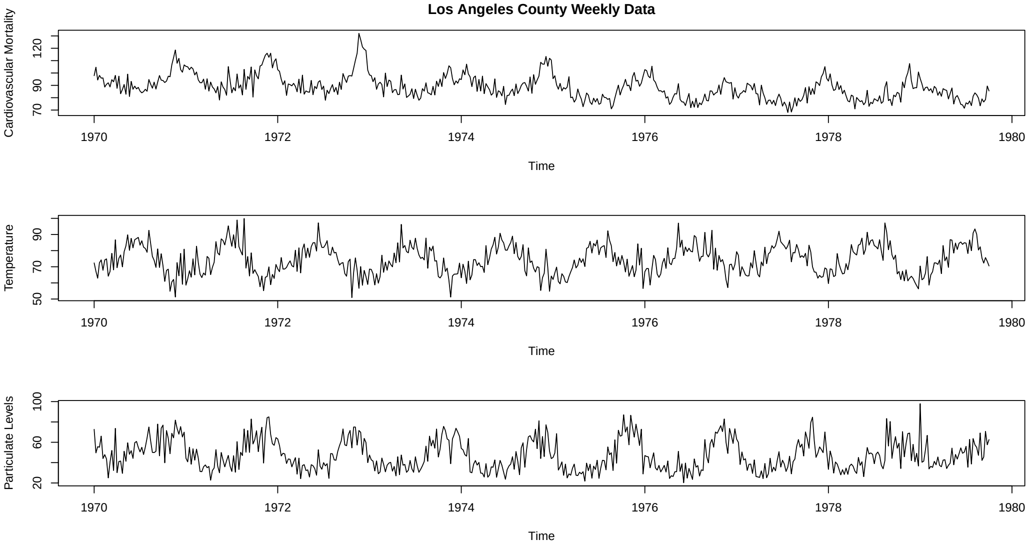
$$\mathbf{y} = B\mathbf{x} + \mathbf{w}_t$$

  and estimate the parameters as previously seen.

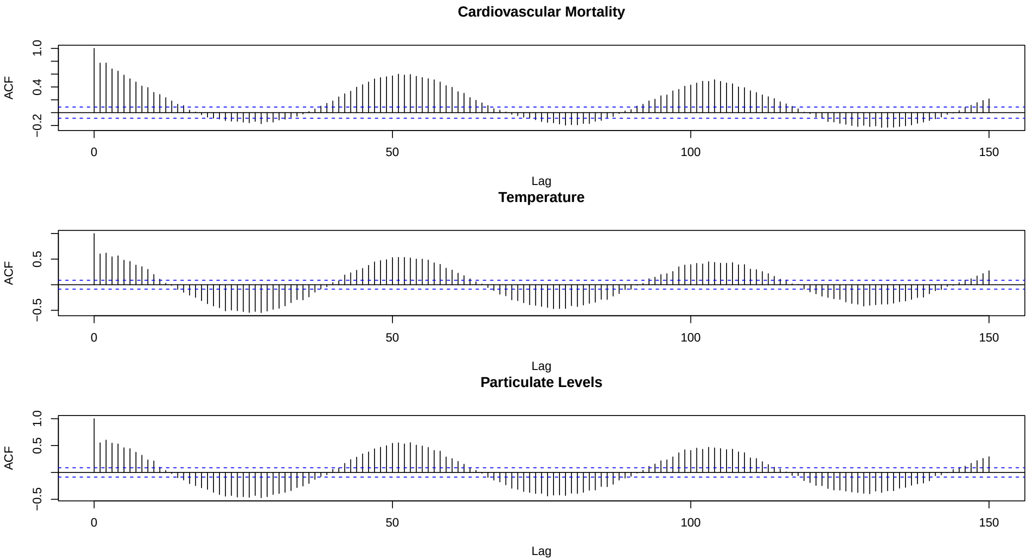
- Estimated covariance matrix of residuals:

$$\hat{\Sigma}_w = \frac{1}{n-1} \sum_{t=2}^{n} (\mathbf{z}_t - \hat{\boldsymbol{\alpha}} - \hat{\Phi}\mathbf{z}_{t-1})(\mathbf{z}_t - \hat{\boldsymbol{\alpha}} - \hat{\Phi}\mathbf{z}_{t-1})'$$

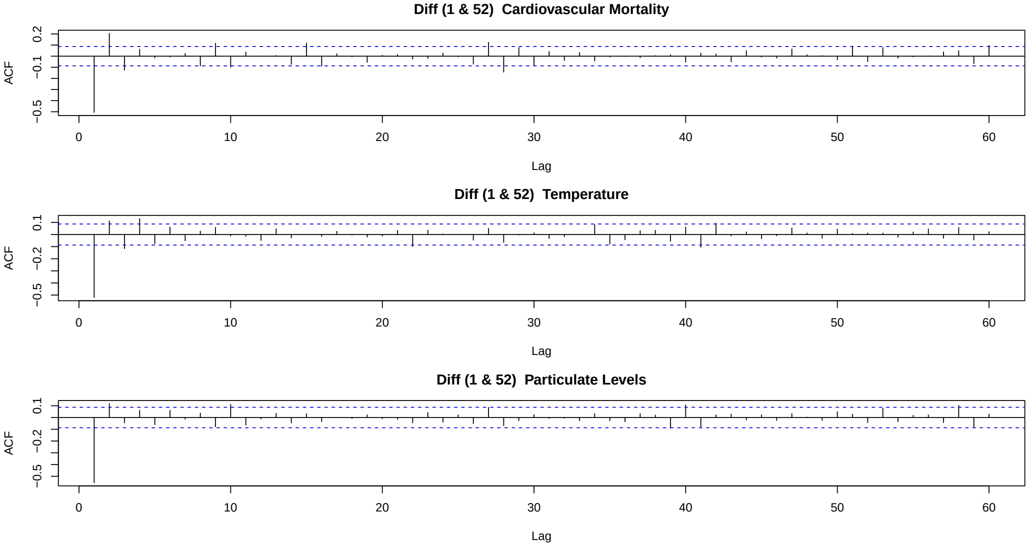
Example: Pollution, Weather, and Mortality



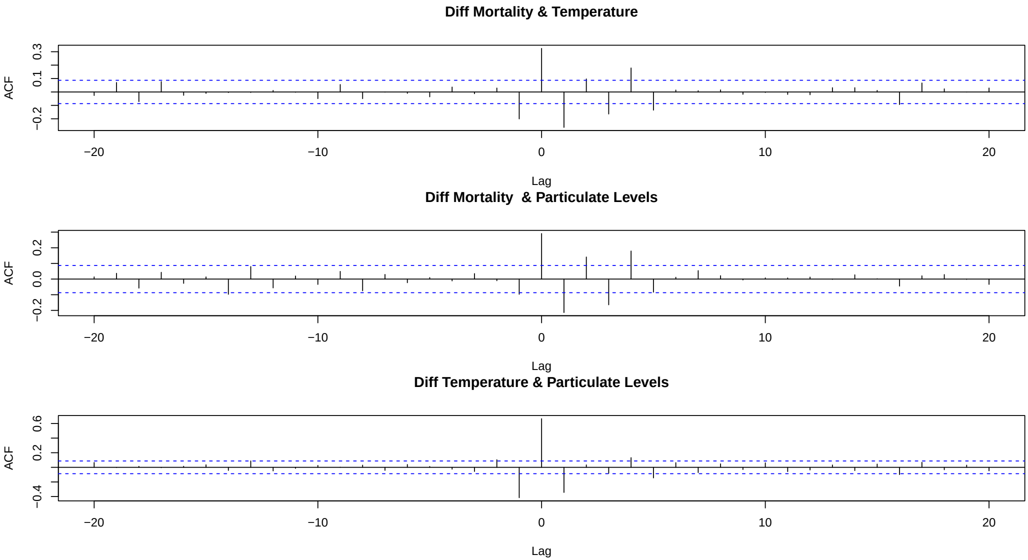
Example: Pollution, Weather, and Mortality



Example: Pollution, Weather, and Mortality



Example: Pollution, Weather, and Mortality



Data Description: Mortality, Temperature, and Pollution

- The data consist of weekly observations over a 10-year period (1970 - 1979) for:
  - Cardiovascular Mortality in Los Angeles County
  - Temperature (weekly averages)
  - Particulate Pollution (e.g., PM levels)

- Strong seasonal components are visible across all three series, driven by winter-summer cycles.

- Cardiovascular mortality shows a downward trend over time.

Example: Pollution, Weather, and Mortality

- Setting: We model the original data without preprocessing for illustrative purpose. We obtain, thus, a three-dimensional time series $\mathbf{x}_t = (x_{t1}, x_{t2}, x_{t3})^\top$, where:
  - x_{t1}: Cardiovascular mortality (M_t)
  - x_{t2}: Temperature (T_t)
  - x_{t3}: Particulate level (P_t)
- We assume a first-order VAR model with trend:

$$
\begin{aligned}
x_{t1} &= \alpha_1 + \beta_1 t + \phi_{11} x_{t-1,1} + \phi_{12} x_{t-1,2} + \phi_{13} x_{t-1,3} + w_{t1} \\
x_{t2} &= \alpha_2 + \beta_2 t + \phi_{21} x_{t-1,1} + \phi_{22} x_{t-1,2} + \phi_{23} x_{t-1,3} + w_{t2} \\
x_{t3} &= \alpha_3 + \beta_3 t + \phi_{31} x_{t-1,1} + \phi_{32} x_{t-1,2} + \phi_{33} x_{t-1,3} + w_{t3}
\end{aligned}
$$

Or, in compact form:

$$\mathbf{x}_t = \boldsymbol{\alpha} + \boldsymbol{\beta} t + \Phi \mathbf{x}_{t-1} + \mathbf{w}_t$$

Fitting the $VAR(1)$ in R

```r
library(vars)
x = cbind(cmort, tempr, part)
summary(VAR(x, p=1, type="both"))
```

```
Estimation results for equation cmort:
======================================
cmort = cmort.l1 + tempr.l1 + part.l1 + const + trend

          Estimate Std. Error t value Pr(>|t|)
cmort.l1  0.464824   0.036729  12.656  < 2e-16 ***
tempr.l1 -0.360888   0.032188 -11.212  < 2e-16 ***
part.l1   0.099415   0.019178   5.184 3.16e-07 ***
const    73.227292   4.834004  15.148  < 2e-16 ***
trend    -0.014459   0.001978  -7.308 1.07e-12 ***
---
```

Estimated Parameters and Covariance

Estimated coefficients:

$$\hat{\boldsymbol{\beta}} = (-0.014, -0.007, -0.005)^\top$$

$$\hat{\boldsymbol{\alpha}} = (73.23, 67.59, 67.46)^\top$$

$$\hat{\Phi} = \begin{bmatrix} 0.46 & -0.36 & 0.10 \\ -0.24 & 0.49 & -0.13 \\ -0.12 & -0.48 & 0.58 \end{bmatrix}$$

Residual covariance matrix:

$$\hat{\Sigma}_w = \begin{bmatrix} 31.17 & 5.98 & 16.65 \\ 5.98 & 40.97 & 42.32 \\ 16.65 & 42.32 & 144.26 \end{bmatrix}$$

Mortality Prediction Equation

From the estimated model, the predicted mortality equation is:

$$\hat{M}_t = 73.23 - 0.014t + 0.46M_{t-1} - 0.36T_{t-1} + 0.10P_{t-1}$$

Model Performance:

- $R^2 \approx 0.69$, indicating 69% of variation in mortality is explained
- Provides interpretable insights into how weather and pollution influence mortality

Graphical Models for Time Series

Undirected Graphs

- A graph $G = (V, E)$ consists of:
  - A set of vertices $V = \{1, \ldots, k\}$.
  - A set of undirected edges $E \subseteq V \times V$ with $(a, b) \in E \Rightarrow (b, a) \in E$.

- In our context, G will model the conditional uncorrelatedness among time series components at the same time point.

- Goal: Construct a graph $G = (V, E)$ where each vertex corresponds to a component of $\mathbf{X}(t)$,

- and:

$$(a, b) \notin E \iff X_a(t) \perp X_b(t) \mid \{X_j(t) : j \neq a, b\} \text{ (linearly)}$$

Time Series Graph

- Each vertex $a \in V$ corresponds to one component $X_a(t)$ of the multivariate process.
- An edge between two nodes a and b indicates that their time series are conditionally correlated, after adjusting for the influence of all other series.
- The absence of an edge (a, b) means that, given all other variables at time t, the components $X_a(t)$ and $X_b(t)$ are linearly conditionally uncorrelated.
- In the Gaussian case, this further implies conditional independence.

Time Series Graph: Why This Matters

- This allows us to identify direct contemporaneous relationships between series.
- Indirect or spurious correlations (e.g., due to shared common drivers) are removed by conditioning on the remaining series.

Multivariate Partial Correlation

To understand the direct linear relationship between two components $X_{a,t}$ and $X_{b,t}$, we aim to remove the influence of all other components.

- Step 1: Define the conditioning set.

$$Y_{ab}(t) = \{X_{j,t} : j \neq a, b\}$$

  This is the set of all time series components except $X_{a,t}$ and $X_{b,t}$.
- Step 2: Remove linear effects. We regress $X_{a,t}$ and $X_{b,t}$ on the entire history of $Y_{ab}(t)$, and define their residuals to be: $\varepsilon_{a,t}$ and $\varepsilon_{b,t}$.
- Step 3 : Compute the Partial Covariance. We calculate the cross autocovariance between the residuals:

$$\gamma^{(ab)}(h) = \mathrm{Cov}(\varepsilon_{a,t}, \varepsilon_{b,t+h})$$

Multivariate Partial Correlation

Interpretation:

- The residuals $\varepsilon_{a,t}$ and $\varepsilon_{b,t}$ represent the part of each process not linearly predictable from the others $\{X_{j,t} : j \neq a, b\}$.
- Any remaining correlation between $\varepsilon_{a,t}$ and $\varepsilon_{b,t}$ indicates a direct linear dependence between $X_{a,t}$ and $X_{b,t}$, not explained by other components.

Partial Correlation Graph

Partial Correlation Graph

Definition: Let $\{\mathbf{X}_t\}$ be a multivariate stationary time series. The partial correlation graph $G = (V, E)$ is defined by:

$$(a, b) \notin E \iff \varepsilon_{a,t} \perp \varepsilon_{b,t} \iff \gamma^{(ab)}(h) = 0 \quad \text{for all } h$$

Key Ideas:

- The residuals isolate the part of each time series not explainable by the rest of the system.
- If these residuals are uncorrelated, there is no direct linear association between the original series after accounting for other series.
- An edge in the graph represents a direct linear dependency.

Special Case: Gaussian Processes

- If $\{\mathbf{X}\}_t$ is Gaussian, the residuals are also Gaussian.
- Then, zero correlation implies independence.
- Hence, the absence of an edge encodes conditional independence.

Example: Conditional Correlation Structure

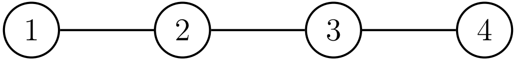
Let $\{\mathbf{X}\}_t$ be defined as:

$$\begin{aligned}
X_{1,t} &= a_1 X_{1,t-1} + \varepsilon_{1,t}, \\
X_{2,t} &= a_2 X_{2,t-1} + b_2 X_{1,t-t_2} + \varepsilon_{2,t}, \\
X_{3,t} &= a_3 X_{3,t-1} + b_3 X_{2,t-t_3} + \varepsilon_{3,t}, \\
X_{4,t} &= a_4 X_{4,t-1} + b_4 X_{3,t-t_4} + \varepsilon_{4,t},
\end{aligned}$$

with some time lags $t_j \in \mathbb{N}$. The noise terms $\varepsilon_{j,t}$ are assumed to be i.i.d. $N(0, \sigma^2)$.

Interpretation:

- Each process $X_{j,t}$ depends on its own past and the past of its predecessor $X_{j-1,t-t_j}$.
- All processes are thus marginally correlated.
- However, in the conditional correlation graph, the structure is:



Partial Spectral Coherence

Partial Spectral Coherence and Graph Structure

Key Idea: Characterize edges in a partial correlation graph using frequency-domain quantities.

Covariance and Spectral Representation:

- Let $\gamma_{ab}(h) = \mathrm{Cov}(X_{a,t+h}, X_{b,t})$.
- The (cross-) spectrum is given by:

$$S_{ab}(\omega) = \sum_{h \in \mathbb{Z}} \gamma_{ab}(h) e^{-2\pi i \omega h}$$

- Collectively, $\mathbf{S}(\omega) = [S_{ab}(\omega)]$ is the spectral density matrix.

Interpretation:

- $S_{ab}(\omega) = 0$ for all $\omega \iff \mathrm{X}_{a,t}$ and $X_{b,t}$ are uncorrelated at all lags.
- The phase of $S_{ab}(\omega)$ reflects time delays at different frequencies.

Partial Spectral Coherence

Removing Linear Effects:

- Define the residuals $\varepsilon_{a,t}$ and $\varepsilon_{b,t}$ as before.
- Let $S_{\varepsilon_a\varepsilon_b}(\omega)$ denote the partial cross-spectrum. This can be obtained using the Spectral matrix as follows:

$$S_{\varepsilon_a\varepsilon_b}(\omega) = S_{ab|\,Y_{ab}}(\omega) = S_{ab}(\omega) - S_{aY}(\omega)S_{YY}(\omega)^{-1}S_{Yb}(\omega)$$

Partial Spectral Coherence:

$$R_{ab|\,Y_{ab}}(\omega) = \frac{S_{ab|\,Y_{ab}}(\omega)}{\sqrt{S_{aa|\,Y_{ab}}(\omega)S_{bb|\,Y_{ab}}(\omega)}}$$

Partial Coherence Graph

Since

$$\varepsilon_{a,t} \perp \varepsilon_{b,t} \iff \gamma^{(ab)}(h) = 0 \quad \text{for all} \;\; h$$

and

$$S_{\varepsilon_a \varepsilon_b}(\omega) = \sum_{h \in \mathbb{Z}} \gamma^{(ab)}(h) e^{-2\pi i \omega h}$$

we obtain the following result:

Proposition. Suppose $G = (V, E)$ is a partial correlation graph for a multivariate time series. Then,

$$(a, b) \notin E \quad \text{if and only if} \quad R_{ab|Y_{ab}}(\omega) = 0 \quad \text{for all } \omega.$$

This provides a frequency-domain criterion for edge absence.

Graph Identification via Spectral Matrix Inversion

Theorem: Let $g(\omega) = \mathbf{S}(\omega)^{-1}$, the inverse spectral matrix. Define:

$$d_{ab}(\omega) = \frac{g_{ab}(\omega)}{\sqrt{g_{aa}(\omega) g_{bb}(\omega)}}$$

Then:

$$d_{ab}(\omega) = -R_{ab|Y_{ab}}(\omega)$$

Implication: The partial spectral coherence can be directly obtained from the normalized inverse spectral matrix.

Key Consequence:

- Edge Detection: Missing edges in the partial correlation graph correspond to zeros in the rescaled inverse spectral matrix:

$$(a, b) \notin E \iff d_{ab}(\omega) = 0 \quad \forall \omega$$

- This mirrors the interpretation of zeros in the concentration matrix (inverse covariance) in Gaussian graphical models.

Practical Implications:

- Estimate $S_{XX}(\omega)$ using periodograms or smoothed estimators.
- Invert $S_{XX}(\omega)$ to obtain $g(\omega)$.
- Check which $g_{ab}(\omega)$ are approximately zero across frequencies.
- Build the edge set E accordingly.

Summary:

- The rescaled inverse spectral matrix provides a practical and interpretable route to identify conditional dependencies.
- This approach bridges graphical models with spectral analysis, offering both theoretical insights and computational advantages.

Frequency-Dependent Connectivity:

- In practice, it may be that $d_{ab}(\omega)$ vanishes only over certain frequency bands.
- This motivates the idea of a frequency-dependent graph, where connections vary across the spectral domain.